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Lecture Note on:

Cesarean Delivery (CD) and Trial of Labor after Cesarean Delivery (TOLAC)

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Contents

Objectives

- At the end of this lecture you will be able to:
  1. Narrate the history of cesarean delivery
  2. Enumerate the regional, national and glonal incidences of cesarean delivery
  3. List the indications and complicaitons of cesarean delivery
  4. Outline the techniques of cesarean delivery

Terminologies

Definition

- **Cesarean Delivery / Cesarean Birth**
  - ✓ Birth of a fetus from the uterus through an abdominal incision
  - ✓ Cesrean section ➔ Not a preferred term
- **Primary Cesarean Delivery**
  - ✓ CD done in a woman without a prior cesarean birth
- **Repeat Cesarean Delivery**
  - ✓ CD done in a woman who had a cesarean birth in a previous pregnancy

- TOLAC
  - ✓ **An attempt** to deliver vaginally with a background of previous cesarean scar regardless of the outcome
- VBAC
  - ✓ **Successful vaginal delivery** after trail of labor with background of previous cesarean scar

- **Scar Dehiscence**
  - ✓Partial thickness
  - ✓Often asymptomatic
- **Uterin Rupture**
  - ✓Full thickness uterine tear
  - ✓Often catastrophic

History

History of Cesarean Delivery

- Evoluation of the term Cesarean
  - Debated over time
    1. Birth of Julius Caesar?

✓History has it that **his mother Aurelia** knows about his success as the greatest general of the that time

✓Women delivered via cesarean delivery that time do not survive the procedure

✓In Caesar's time CD was reserved for women who were **dead or dying**

History Cont'd

2. Romna Law
   - ✓ Under Numa Pompilius the first ("Lex Regia"), then renamed after Caesar ("Lex Cesarea") specified surgical removal of the fetus before burial of the deceased pregnant woman; religious edicts required separate burial for the infant and mother.
3. The term *cesarean* may also refer to patients being cut open, because the Latin verb ***caedare*** means to cut.

History Cont'd

- For hundreds of years CD was associated with high morbidity and mortality
  - Primitive technique
  - Lack of antispesis
  - Hemorrhage (surgeons attempted to leave the uterus open fearing that the sutures would serve as a nidus for infection)

History cont'd

- 1598
  - ✓Guillimeau introduced the term "Section"
  - ✓Before his time the term in use was "Cesarean Operation"
- 1769
  - ✓ Lebas started advocating suturing the uterus
- 1876
  - ✓Eduardo Porro advocated for supracervical hysterectomy and bilateral salpingo-oopherectomy during CD to control bleeding and to prevent postoperative infection.

- Shortly thereafter, surgeons gained experience with internal suturing because **silver-wire stitches** were developed by the gynecologist **J. Marion Sims**, who had perfected the use of these sutures in the treatment of vesicovaginal fistulae resulting from obstructed labors.

- In the early 1880s, two German obstetricians, **Ferdinand Adolf Kehrer** (1837-1914) and **Max Sänger** (1853-1903), both independently proposed a **transverse incision** of the lower segment of the uterus, at the level of the internal cervical os, and developed two-layer uterine closure methods with the sutures used by J. Marion Sims.
- Another pivotal contribution was made in 1900 by **Hermann Johannes Pfannenstiel (1862-1909),** a German gynecologist who described a transverse suprapubic incision, or pelvic skin incision.

- Between 1890 and 1925, more and more surgeons began using transverse incisions of the uterus.
- **John Martin Munro Kerr** (1868-1960), a professor of Obstetrics Midwifery at the University of Glasgow, popularized the Pfannenstiel skin incision and lower segment uterine incision and is considered the “father” of the modern CD.

- It was noted that such incisions reduced the rate of infection and the risks of incisional hernia and rupture with subsequent pregnancies compared with the vertical incisions
- However, before the advent of antibiotics, owing to the risk for peritonitis, **extraperitoneal cesarean** was advocated by Frank (1907), Veit and Fromme (1907), and Latzko (1909) and was popularized by Beck (1919) in the United States.

- Interestingly, the vertical opening of the abdomen was still the main technique used in the 1970s, although it was known from the beginning of the twentieth century to be associated with higher rates of long-term postoperative complications, such as:
  - ✓ wound dehiscence and abdominal incision hernia, and
  - ✓ it is also cosmetically less pleasing.

- **The introduction of penicillin in 1940 dramatically reduced the risk for peripartum infections. As antibiotic therapy emerged, the need for extraperitoneal dissection diminished.**
- As technology developed, including improved anesthesia, and the medical management of pregnancy and childbirth accelerated, CD became more commonplace in obstetrics.
- Given its current safety and effectiveness, a liberalized approach to using cesarean childbirth has emerged in developed countries over the past 40 years.

Incidence

- The CD *rate* describes the proportion of women undergoing CD of all women giving birth during a specific time period.
- The CD rate may be further subdivided into ***primary and repeat CD rates*** both as a proportion of the entire obstetric population
- **CD rates have risen in the United States in a dramatic fashion from less than 5% in the 1960s to 32.7% by 2013, with stable rates around 32% to 33% in the last 5 years or so**

- CD accounts for more than 1 million major operations performed annually in the United States.
- It is the most common major surgical procedure undertaken today in the United States and around the world.

- Among the reasons for this increase are
  - ✓a continued increase in primary CDs for dystocia, failed induction, and malpresentation;
  - ✓an increase in the proportion of women with obesity, diabetes mellitus, and multiple gestation, which predispose to CD;
  - ✓Increased practice of CD on maternal request; and
  - ✓Limited use of a trial of labor after cesarean (TOLAC) delivery due to both safety and medicolegal concerns.

- A recent increase in international CD rates has also been documented
  - ✓ Rates of about 25% to 30% are reported in some European countries, such as the United Kingdom;
  - ✓ They are over 40% in Italy,
  - ✓ Over 50% in China, and
  - ✓ Even higher in places like Brazil and Egypt

- **The World Health Organization (WHO) has proposed an incidence of CD between 10% and 15% as a target to optimize maternal and perinatal health.**
- However, it is not possible to determine an optimal CD rate because any ideal rate must be a **function of multiple clinical factors** that vary in each population and are influenced by the level of obstetric care provided

- Hence, although **CD rates** can be considered a measure of a specific health care process (mode of delivery), these rates **are not appropriate outcome measures** because they do not indicate whether cesarean or vaginal birth results in optimal perinatal outcomes.
- **Maternal and perinatal morbidity and mortality should be the outcomes monitored to ensure best quality of care.**
- Higher CD rates (e.g., **15% to 20%** compared with **<5% to 10%**) have been associated with better perinatal outcomes in several studies

- So **instead of setting goals or limits for overall CD rates, it is most important to monitor maternal and perinatal health outcomes**
-

Factors responsible for increased cesarean rates

Obstetric Factors

1. Increased primary CD rate
2. Failed induction, increased use of induction
3. Decreased use of operative vaginal delivery
4. Increased macrosomia, CD for macrosomia
5. Decline in vaginal breech delivery
6. Increased repeat CD rate
7. Decreased use of vaginal birth after CD

Maternal Factors

1. Increased proportion of women >35 yr
2. Increased proportion of NP women
3. Increased primary CDs on maternal request

Physician Factors

1. Malpractice litigation concerns

Safe Prevention of CD

First Stage of Labor

• A prolonged latent phase (>20 hr in NP women and >14 hr in MP women) should not be an indication for CD. (Grade 1B)

• Slow but progressive labor in the first stage rarely should be an indication for CD. (Grade 1B)

• As long as fetal and maternal status are reassuring, cervical dilation of 6 cm should be considered the threshold for the active phase in most laboring women. Thus before 6 cm of dilation is achieved, standards of
active-phase progress should not be applied. (Grade 1B)

• CD for active-phase arrest in the first stage of labor should be reserved for women at or beyond 6 cm of dilation with ruptured membranes who fail to progress despite 4 hr of adequate uterine activity or at least 6 hr of oxytocin administration with inadequate uterine activity
and no cervical change. (Grade 1B)

Safe Prevention Cont'd

Second Stage of Labor

• A specific absolute maximum length of the second stage of labor above which all women should be delivered operatively has not been identified. (Grade 1C)

• Before diagnosing arrest of labor in the second stage, if the maternal and fetal conditions permit, allow for the following:

• At least 2 hr of pushing in MP women (Grade 1B)

• At least 3 hr of pushing in NP women (Grade 1B)

Longer durations may be appropriate on an individualized basis (e.g., with the use of epidural analgesia or with fetal malposition) as long as progress is being documented. (Grade 1B)

• Operative vaginal delivery in the second stage of labor should be considered an acceptable alternative to CD. Training in, and ongoing maintenance of, practical skills related to operative vaginal delivery should be encouraged. (Grade 1B)

• Manual rotation of the fetal occiput in the setting of fetal malposition in the second stage of labor is a reasonable alternative to operative vaginal delivery or CD. To safely prevent CD in the setting of malposition, it is important to assess fetal position throughout the second stage of labor. (Grade 1B)

Safe Prevention Cont'd

Fetal Heart Rate Monitoring

• Amnioinfusion for repetitive variable fetal heart rate decelerations may safely reduce the CD rate. (Grade 1A)

• Scalp stimulation can be used as a means of assessing fetal acid-base status when abnormal or indeterminate (*non – reassuring*) fetal heart patterns (e.g., minimal variability) are present, and it is a safe alternative to CD in this setting. (Grade 1C)

Safe Prevention Cont'd

Induction of Labor

• Induction of labor generally should be performed based on maternal and fetal medical indications and after informed consent is obtained and documented. Inductions at 41 0/7 weeks of gestation and beyond should be performed to reduce the risk of CD and the risk of perinatal morbidity and mortality. (Grade 1A)

• Cervical ripening methods should be used when labor is induced in women with an unfavorable cervix. (Grade 1B)

• If the maternal and fetal status allow, CDs for failed induction of labor in the latent phase can be avoided by allowing longer durations of the latent phase (up to 24 hr or longer) and requiring that oxytocin be administered for at least 18 hr after membrane rupture before deeming the induction a failure. (Grade 1B)

Safe Prevention Cont'd

Fetal Malpresentation

• Fetal presentation should be assessed and documented beginning at 36 0/7 weeks of gestation to allow for external cephalic version to be offered. (Grade 1C)

Suspected Fetal Macrosomia

• CD to avoid potential birth trauma should be limited to estimated fetal weights of at least 5000 g in women without diabetes and at least 4500 g in women with diabetes. The prevalence of birthweight of 5000 g or more is rare, and patients should be counseled that estimates of fetal weight, particularly late in gestation, are imprecise. (Grade 2C)

• Women should be counseled about the IOM maternal weight guidelines in an attempt to avoid excessive weight gain. (Grade 1B)

Safe Prevention Cont'd

Twin Gestations

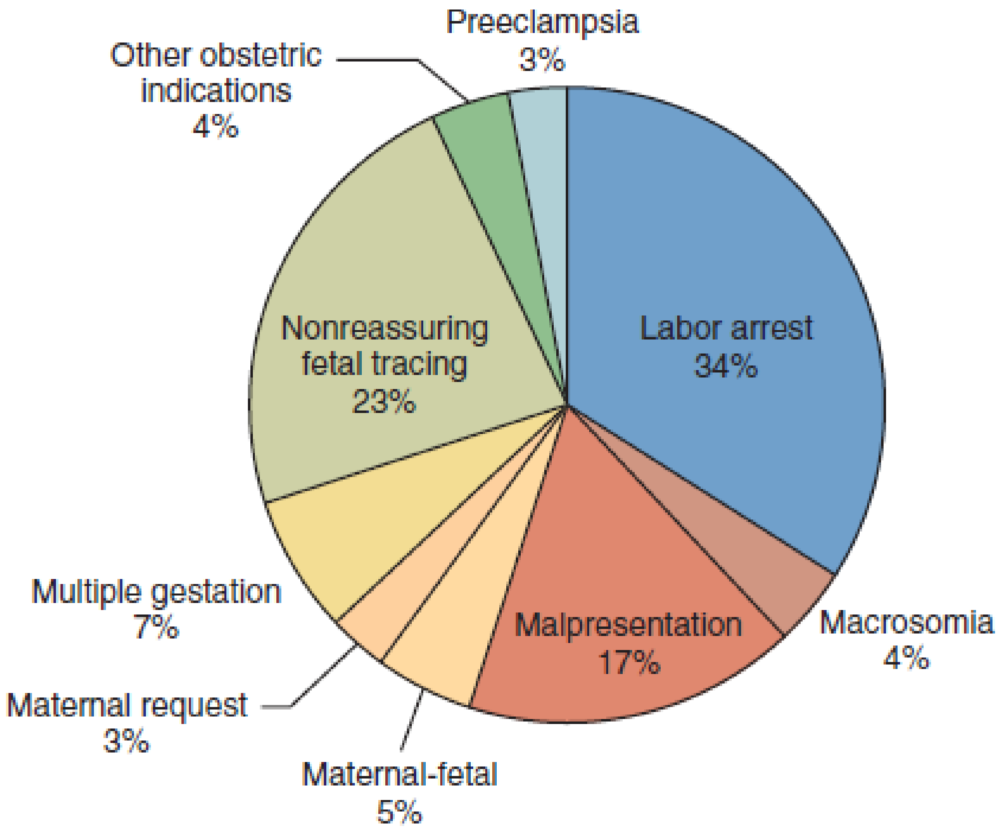
• Perinatal outcomes for twin gestations in which the first twin is in cephalic presentation are not improved by CD. Thus women with either cephalic/cephalic–presenting twins or cephalic/noncephalic-presenting twins should be counseled to attempt vaginal delivery. (Grade 1B)

Other

• Individuals, organizations, and governing bodies should work to ensure that research is conducted to provide a better knowledge base to guide decisions regarding CD and to encourage policy changes that safely lower the rate of primary CD. (Grade 1C)

Indications for Cesarean Delivery

- CD can be performed for maternal-fetal, fetal, and maternal indications.
- The most common current indications are, in order of frequency,
  1. Failure to progress, also called *cephalopelvic disproportion* (CPD) or *dystocia* (about 30%);
  2. Prior cesarean (30%);
  3. Nonreassuring fetal heart rate (FHR) patterns (10%); and
  4. Fetal malpresentation (10%).



Indications for primary cesarean delivery. (Data from Barber EL, Lundsberg LS, Belanger K, Pettker CM, Funai EF, Illuzzi JL. Indications contributing to the increasing cesarean delivery rate. *Obstet Gynecol.* 2011;118:29-38.)

Risks and Benefits of CD without Indications

Potential Benefits

- Reduction in perinatal morbidity and mortality:

✓ Elimination of intrapartum events associated with perinatal asphyxia

✓ Reduction in traumatic birth injuries

✓ Reduction in stillbirth beyond 39 weeks' gestation

- Possible protective effect against pelvic floor dysfunction
- Less postpartum hemorrhage

Risks and Benefits of CD without Indications

Potential Risks

1. Increased short-term maternal morbidity
2. Increased endometritis, transfusion, and venous thrombosis rates
3. Increased length of stay and longer recovery time
4. Increased short-term neonatal morbidity
5. Increased mild neonatal respiratory morbidity
6. Increased long-term maternal and neonatal morbidity
7. Increased risk for placenta accreta and hysterectomy with subsequent cesarean deliveries

Selected Indications by Category

Maternal-Fetal

1. Cephalopelvic disproportion
2. Placental abruption
3. Placenta previa
4. Repeat cesarean delivery
5. Cesarean delivery on maternal request

Maternal

1. Specific cardiac disease (e.g., Marfan syndrome with dilated aortic root)

Fetal

1. Nonreassuring fetal status
2. Breech or transverse lie
3. Maternal herpes

Maternal-Fetal Indications

- Most CDs are performed for conditions that might pose a threat to both mother and fetus if vaginal delivery occurred.
- Complete placenta previa and placental abruption with the potential for hemorrhage are clear examples.
- Dystocia presents a risk for both direct fetal and maternal trauma, and it may also compromise fetal oxygenation and metabolic status.

Fetal Indications

- **Fetal indications are primarily recognized by nonreassuring**
- **FHR testing with the potential for long-term consequences of metabolic acidosis**
- Continuous FHR monitoring is associated with a significant reduction in neonatal seizures and remains the most commonly used modality for fetal monitoring in labor
- Scalp stimulation can be used to ameliorate the high false-positive rate of continuous FHR monitoring

Fetal Indications

- Other fetal indications for CD include malpresentation, such as a breech orientation, and more than 90% of these babies in
- singleton gestations are delivered by cesarean.9 Active maternal herpes infection would be an indication for CD to reduce the risk for transmission of infection

Maternal Indications

- **Maternal indications for CD are relatively few and can be considered as medical or mechanical in nature**
- Certain maternal cardiac conditions, such as a dilated aortic root with Marfan syndrome, are indications for CD
- Central nervous system abnormalities in which increased intracranial pressure would be undesirable, such as accompanies the second stage of labor, have also led to recommendations for CD

Maternal Indications

- Alterations in the capacity of the maternal pelvis can be indications for CD
- Mechanical vaginal obstruction as a result of pelvic masses such as lower segment myomata are examples
- Finally, women with massive condylomata may also require CD, but this is rare

Cesarean Delivery on Maternal Request

- As CD has become safer, women have occasionally voiced their wish for a cesarean *without* a medical indication. This clinical scenario has been recently called “cesarean delivery on maternal request.”
- At times, physicians have also advocated cesarean as the preferred mode of delivery, even in the absence of accepted indications as described previously
- We would term this scenario “cesarean delivery on physician request.”

Techniques of Cesarean Delivery

Precesarean Antibiotics

- **Prophylactic preoperative antibiotics are of clear benefit in reducing the frequency of postcesarean endomyometritis and wound infection in both laboring and nonlaboring CDs**
- Prophylactic antibiotics should be given approximately 30 to 60 minutes before the skin incision to allow for adequate tissue concentrations

Precesarean Thromboprophylaxis

- **Because venous thromboembolism (VTE) is the leading cause of maternal mortality in developed countries, and CD increases this risk, thromboprophylaxis should be considered in all CDs**
- Mechanical prophylaxis with graduated compression stockings or a pneumatic compression device is recommened during and after every CD until ambulation resume

- Women with additional risk factors such as morbid obesity, prior VTE, or immobility may benefit from medical thromboprophylaxis after cesarean (e.g., with prophylactic heparin).

Other Prophylactic Precesarean Interventions

- A lateral tilt of about 15 degrees is suggested to elevate the mother's right side to avoid vena caval compression and supine hypotension syndrome.
- However, left lateral tilt, head-up or head-down position, the use of wedges and cushions, flexion of the table, and use of a mechanical displacer have been insufficiently studied to provide any strong recommendation for routine clinical use

- Vaginal preparation immediately before CD with povidoneiodine solution significantly reduces the incidence of postcesarean endometritis (from 8.3% in those who received placebo preparations to 4.3%), especially in women with ruptured membranes (17.9% reduced to 4.3%) and those in labor (13.0% reduced to 7.4%).

- **Urinary bladder catheterization for CD is prudent until evidence can delineate that eliminating this practice will not result in an increase in bladder or ureteral injury.**
- The drainage tube is inserted when analgesia is established and is then left in situ for 12 to 24 hours until the patient is able to mobilize.
- Indwelling catheter placement in hemodynamically unstable women is recommended to monitor urine output and evaluate fluid balance.

Site Preparation

- Hair removal
  - ✓Clipping of the hair on the morning of surgery
- Surgical scrub
  - ✓Chlorhexidinealcohol scrub has been associated with a lower incidence of wound infection compared with povidone-iodine scrub
- Drapes
  - ✓ Avoid adhesive Drapes

Abdominal Incision

- Factors that influence the type of incision include
  1. the urgency of the delivery,
  2. placental disorders such as anterior complete placenta previa and placenta accreta,
  3. prior incision type, and
  4. the potential need to explore the upper abdomen for nonobstetric pathology

- When a transverse Pfannenstiel skin incision is used, it is made about two fingerbreadths (1 inch or 2.5 cm) above the symphysis in the midline and is extended laterally in a slightly curvilinear manner
- The length of the incision should be based on the estimated fetal size; at term, it usually should be about 15 cm, or the length of an Allis clamp
- The site of the incision, either below or above the pannus or either vertical or transverse, has not been sufficiently studied in obese individuals to provide an evidence-based recommendation.

Abdominal Skin Incision and Abdominal Entry

Transeverse Incision

- Greater cosmotic results
- Less wound dehiscence

Midline (Vertical)

- Faster abdominal entry
- Less bleeding
- Less superficial nerve injury
- Easily extendable

- Occasionally, a transverse incision of the rectus sheath and muscles (***Maylard incision***) is necessary for proper exposure and room to deliver the fetus (e.g., with massive fetal hydrocephaly)
  - ✓ In these cases, only the medial half of the muscle is incised to avoid lacerating the deep epigastric vessels
- Complete transection of the rectus muscles is referred to as the ***Cherney incision,*** which requires identification of the epigastric vessels and ligation bilaterally.

- Following the skin incision, the subcutaneous tissue is then bluntly pushed away to identify the underlying fascia
- In repeat operations, sharp dissection of the subcutaneous adipose tissue may be required
- The fascia is incised and dissected or is bluntly extended in a mild curvilinear manner bilaterally
- It should be tented with the surgeon’s forceps to separate it from the underlying muscle and to identify perforating vessels, which may require ligation or coagulation
- Curvilinear extension is essential because direct transverse extension often leads to inadvertent muscle incisions and bleeding

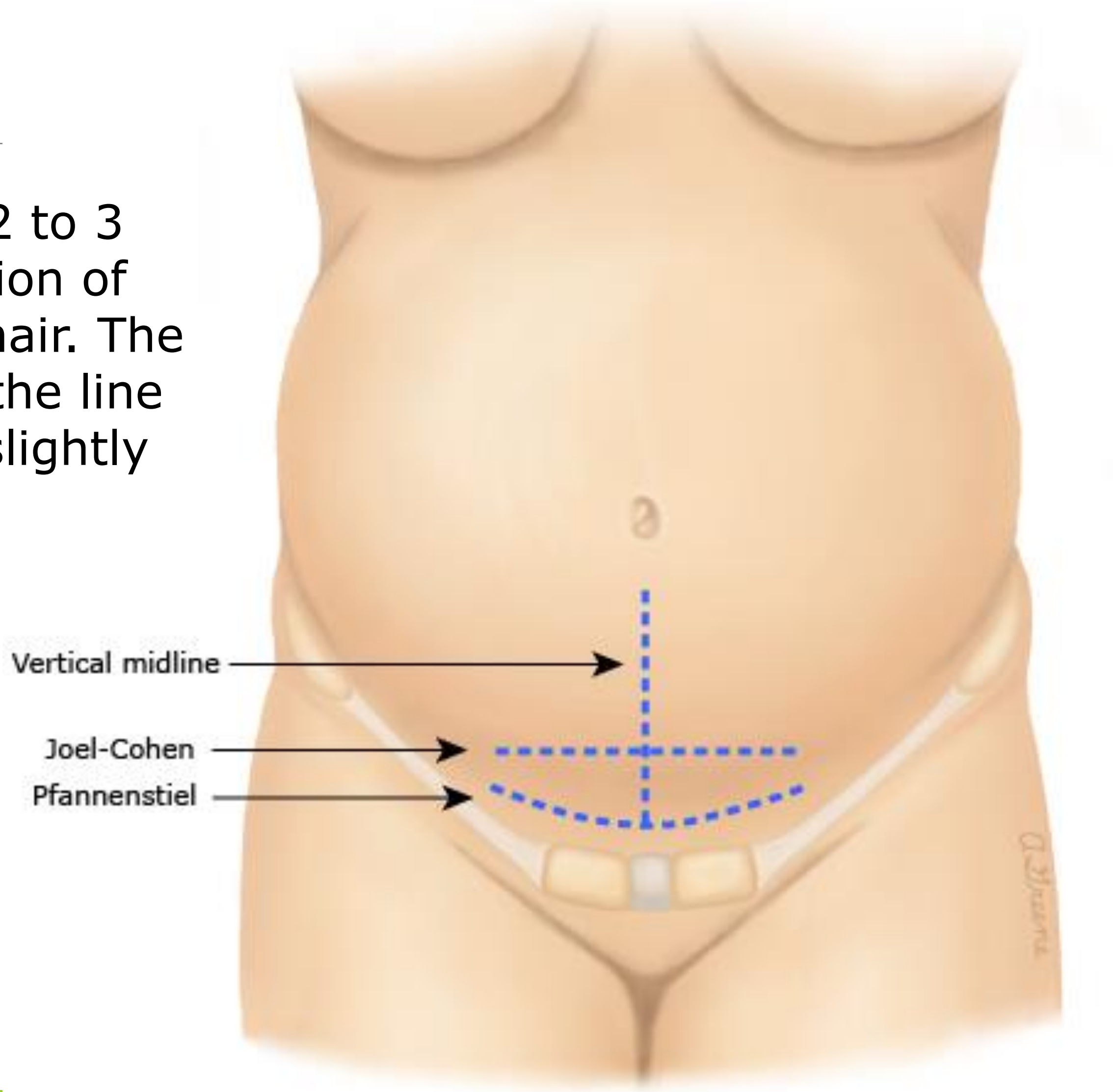
- Once the fascial incision is completed, the fascia is then grasped in the midline bilaterally and is separated from the underlying rectus muscles superiorly and inferiorly by blunt and sharp dissection from the median raphe
- The rectus muscles can be separated bluntly in the midline to reveal the posterior rectus sheath and peritoneum, which can also be entered bluntly with the fingers to avoid trauma to the underlying bowel.
- The point of entry should be as superior as possible to avoid bladder injury, particularly in repeat operations in which the bladder may be adherent superiorly

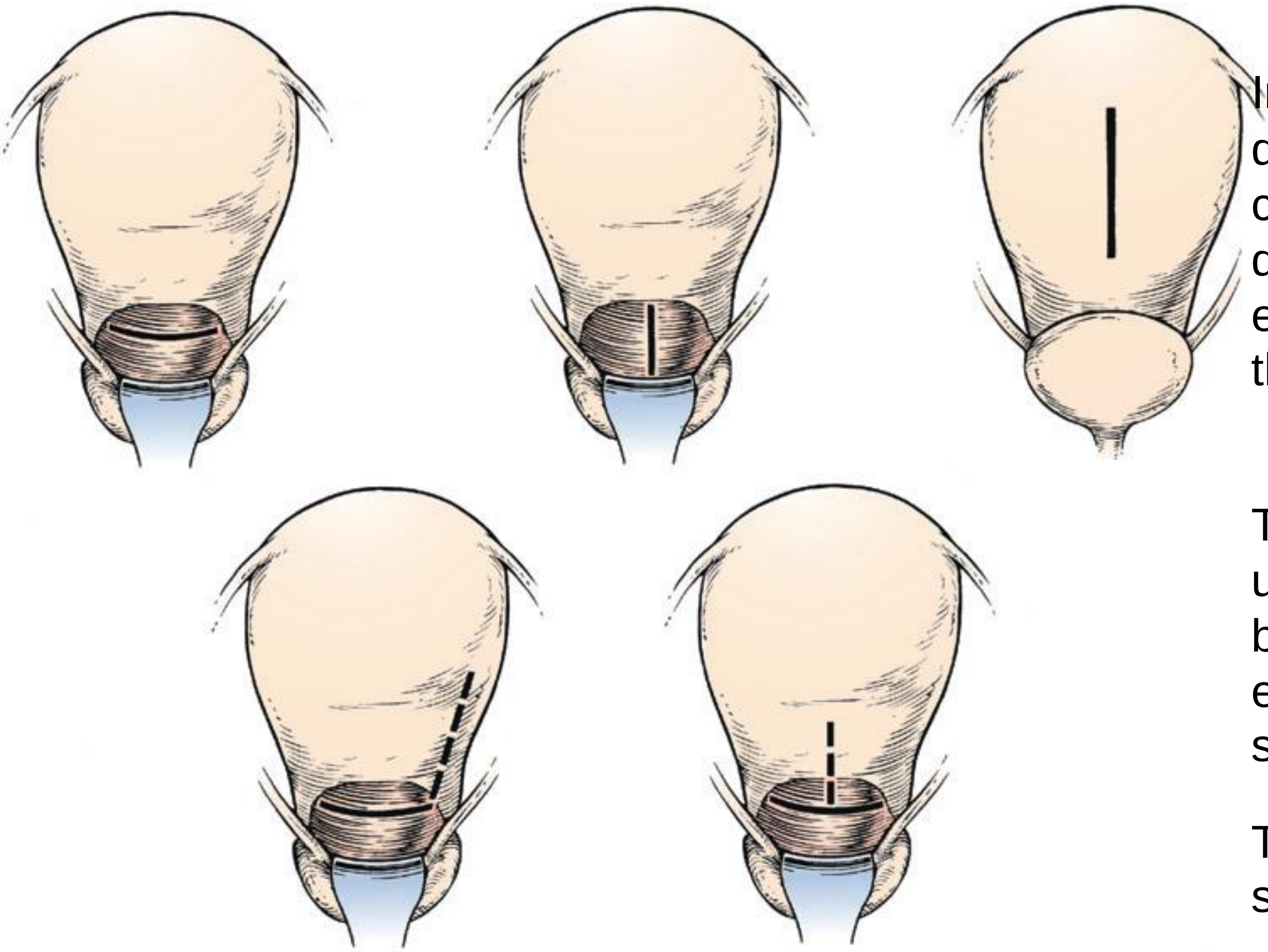
Bladder Flap

- Currently not recommended
  - ✓ Studies show that bladder flap creation increases operation time without improving operative outcome

Abdominal Incision for CD: Types

The Pfannenstiel skin incision is slightly curved, 2 to 3 cm above the symphysis pubis, with the midportion of the incision within the clipped area of the pubic hair. The Joel-Cohen type incision is straight, 3 cm below the line that joins the anterior superior iliac spines, and slightly more cephalad than Pfannenstiel.





In the *low transverse incision* (performed in >90% of cesare deliveries) the incision is made in the lower uterine segment curving gently upward. If the lower segment is poorly developed, the incision can also curve sharply upward at each end to avoid extending into the ascending branches of the uterine arteries

The *low vertical incision* is made vertically in the lower uterine segment, avoiding extension into the bladder below. If more room is needed, the incision can be extended upward into the upper uterine segment

The *classical incision* is entirely within the upper uterine segment and can be at the level shown or in the fundus

With the *J incision,* if more room is needed when an initial transverse incision has been made, either end of the incision can be extended upward into the upper uterine segment and parallel to the ascending branch of the uterine artery.

With the *T incision,* more room can be obtained in a transverse incision by an upward midline extension into the upper uterine segment.

- The low transverse uterine incision replaced the vertical uterine incision at the beginning of the twentieth century

- The low transverse incision is preferred to a vertical incision because it:
  - ✓ is associated with less blood loss,
  - ✓ is easier to perform and repair, and
  - ✓ provides for the option of subsequent TOLAC because the rate of subsequent rupture is lower than with incisions that incorporate the upper uterine segment

- Indications for vertical uterine incision
  - ✓ Lower uterine segment is poorly developed (e.g., at 23 to 25 weeks gestation).
  - ✓Fetus is in a back-down transverse lie.
  - ✓An anterior leiomyoma obstructs the lower uterine segment.
  - ✓ Complete anterior placenta previa or placenta accreta is present
  - ✓ Certain fetal abnormalities such as massive hydrocephalus, a very large sacrococcygeal teratoma, or to deliver conjoined twins.

Delivery of the fetus

- Cephalic
- Reverse breech
- Push
- DCC or umblical cord milking for those less than 37 weeks of GA

Prevention of Postpartum Hemorrhage

- **Studies suggest that 10 to 80 IU of oxytocin in 1 L crystalloid infused over 4 to 8 hours significantly prevents uterine atony and postpartum hemorrhage**

Placental Extraction

- Spontaneous expulsion with gentle cord traction has been shown by several RCTs to be associated with less blood loss and a lower rate of endometritis than manual extraction

Uterine Repair

- Exterorize the uterus
- Control bleeding from the edges with clamps
- Manually curette with moistened sponge
- Close in 2 layers

Abdominal Closure

- Layer by layer
  - ✓ Controversies
    - Closing the peritoneum Vs Leaving it open
    - Approximating the muscles Vs leaving it open
    - Skin closure: Staples Vs Suture Closure

Complications of Cesarean Delivery

- Uterine lacerations
- Bladder injury
- Ureteral Injury
- Gastrointestinal tract injury
- Uterine atony
- Risk of placental previa and placental accrete
- Maternal mortality

TOLAC

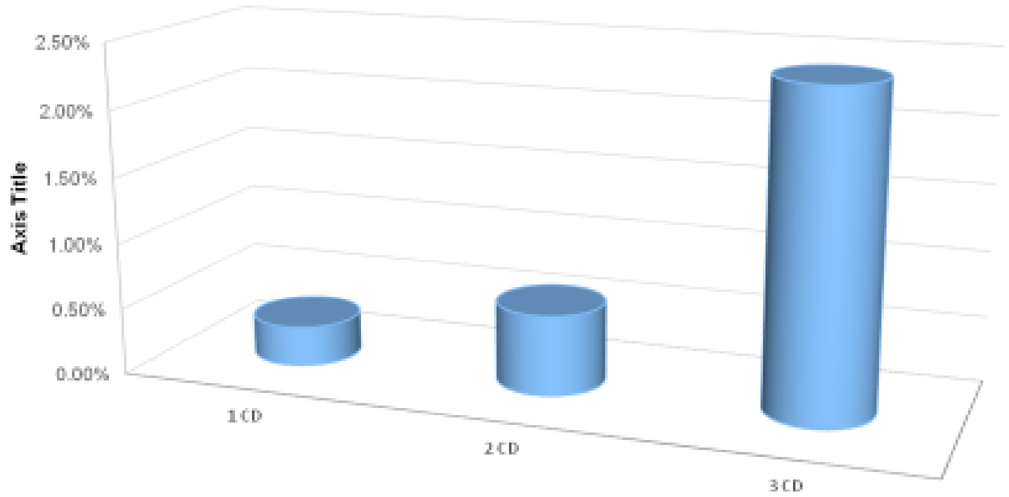
Benefits of TOLAC

- VBAC
  - ✓ Avoids major abdominal surgery
  - ✓ Decreased maternal morbidity; lower rates of:
    - Hemorrhage
    - Thromboembolisim
    - Infection
    - Shorter recovery period

Benefits Cont'd

- ✓ Decreased risk of complications in future pregnancies; for those considering future pregnancies, less risks of:
  - Hysterectomy
  - Bowel or bladder injury
  - Transfusion
  - Infection
  - Accreta syndrome
- ✓ Decreased overall cesarean delivery rate

Risk of Placenta Accreta Syndrome - In the absence of PP



	1 CD	2 CD	3 CD
Y-Values	0.30%	0.60%	2.40%

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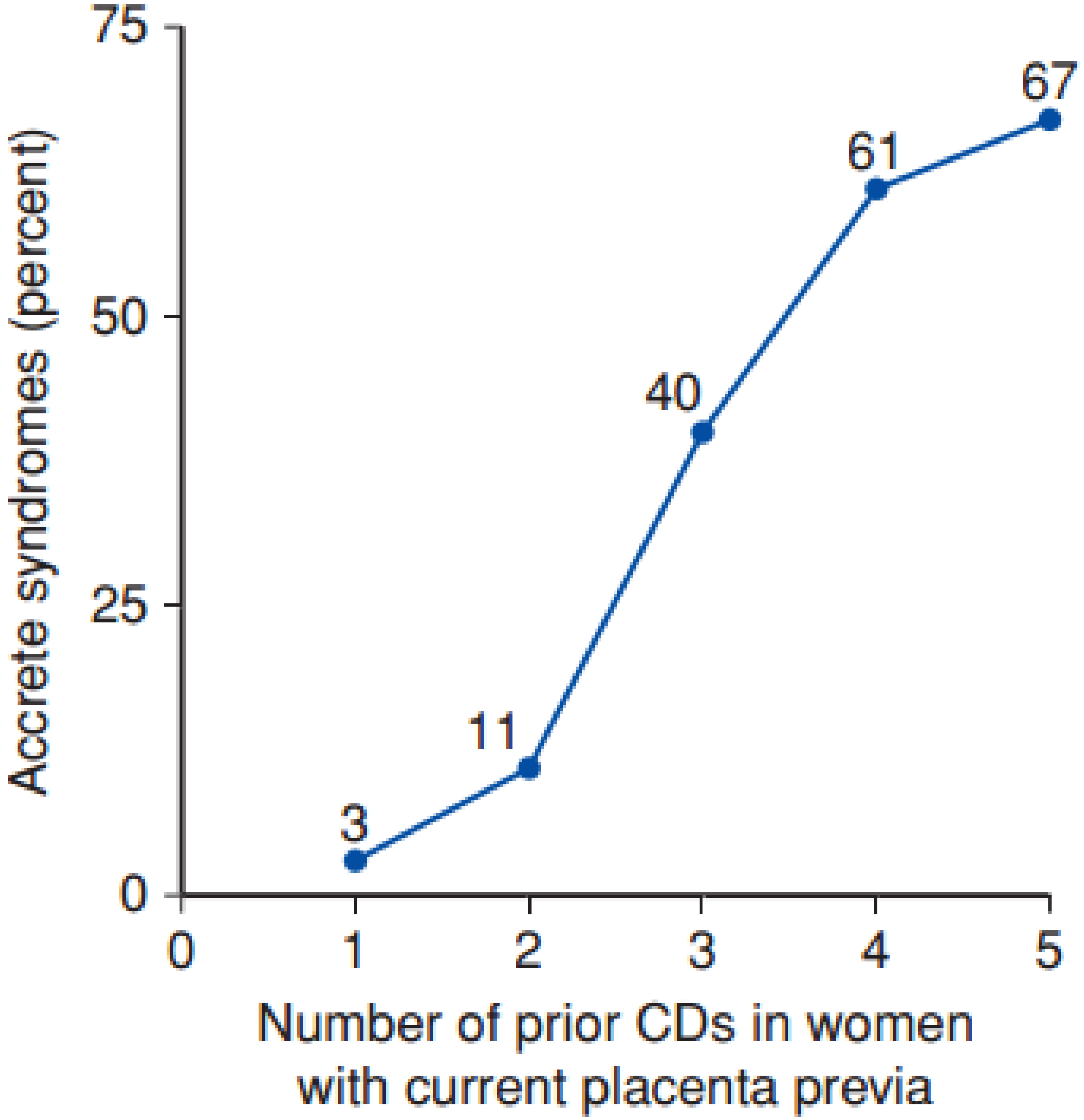
75
50
25
0
Accrete syndromes (percent)
3
11
40
61
67
0
1
2
3
4
5
Number of prior CDs in women
with current placenta previa

Disadvantages of TOLAC

- Failed TOLAC
  - ✓Increased maternal and perinatal morbidity when compared with successful VBAC or elective repeat cesarean delivery
  - ✓Most maternal morbidity related to TOLAC occurs when repeat cesarean delivery becomes necessary

Candidates for TOLAC

- 1 previous low uterine segment cesarean scars
- 1 previous unknown type of uterine scar unless the previous cesarean delivery was performed at an extremely preterm gestational age
- Twin pregnancy with 1 previous lower uterine segment uterine cesarean scar
- Documented prior low vertical uterine incision that does not involve the contractile segment of the uterus
- Estimated fetal weight < 4 K.g.

Candidates cont'd

- Spontaneous onset of labor
- Operation theatre immediately available
- Cephalic fetal presentation
- Mother should understand the risks and benefits of TOLAC and give a verbal consent to her physician
- It is reasonable to consider women with two previous LUST CDs to be candidates for TOLAC and to counsel them based on the combination of other factors that affect their probability of achieving a successful VBAC.

Contraindications for TOLAC

- History of uterine rupture
- History of extensive transfundal uterine surgery
- Previous classical or T-incision
- More than 2 previous LUST CS
- An attempt of cervical rippening with prostaglandins for unfavorable cervix

- Most literatures connote 60-80% VBAC success rates in carefully selected women
- However, success of VBAC varies based on the individual partturients demographics and obstetric profile

Factors that favor success

- Women who have history of vaginal delivery (before or after the previous cesarean section)
- Previous indication for nonrecurring conditions like breech presentation
- Labor onset before 40 weeks of gestation
- Normal BMI
- Normal birthweight

Factors that negatively influence success of VBAC

- If previous indciation was for CPD (especially if the current EFW is greater than the previous pregnancy outcome that led to cesarean delivery)
- High maternal BMI
- Advanced gestational age at delivery (gestational age above 40 weeks)
- The presence of preeclampsia at the time of delivery

Factors that negatively influence cont’d

- Shorter interprergancy interval
- High birthweight
- Pregnancy at advanced maternal age
- Physicians should consider past birthweights and current estimated fetal weight when making decisions regarding TOLAC. Suspected macrosomia alone should not preclude offering TOLAC.

The following are not contraindicated for TOLAC

- Epidural or systemic analgesia during trial of labor
- External cephalic version
- Augmentation of labor with oxytocine

Labor curve for TOLAC

- Similar to those who are nulliparous women
- For example: Prolonged second stage for a women on TOLAC and who has history of vaginal delivery must be 2 hours; not 1 hour.
- Follow up should include TOLAC Chart on top of partograph (both TOLAC chart and Partograph should be used simultaneously)

Sings of uterine rupture during TOLAC

- High index of susciption must be maintained during TOLAC follow up
- Fetal heart rate abnormality is the earliest sign in nearly 70% of the cases
- Abdominal and uterine tenderness and vaginal bleeding are th other signs
- Cesesioin of uterine contraction can be the late sign
- In a woman with repaired uterine rupture, subsequent deliveries should be conducted in between 36 weeks + 0 day -38 weeks +6 days
- Risk of rupture if labor starts can go as high as 6% for lower uterine segment reparied scar and upto 32% for fundal uterine rupture repaired scar

Counseling a woman for TOLAC

- Should start early in the ANC course
- As the pregnancy progresses, if other circumstances arise that may change the risks or benefits of TOLAC, these should be adressed
- Counseling may also include
  - ✓ Considerations of family size
  - ✓ Risks of additional cesarean deliveries
  - ✓ Future reproductive plans may be uncertain or change
- Site of delivery should be selected
- Form of family planning should be adressed during ANC and if choices change with time, it has to be documented.

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Thank you for listening!

Extra Notes

Levels of Evidence:

IA	Evidence from meta-analysis of randomized controlled trials
IB	Evidence from at least one randomized controlled trial
IIA	Evidence from at least one controlled study without randomization
IIB	Evidence from at least one other type of quasi-experimental study
III	Evidence from non-experimental descriptive studies, such as comparative studies, correlation studies, and case-control studies
IV	Evidence from expert committee reports or opinions or clinical experience of respected authorities, or both

Grades of Recommendations:

A	Directly based on Level I evidence
B	Directly based on Level II evidence or extrapolated recommendations from Level I evidence
C	Directly based on Level III evidence or extrapolated recommendations from Level I or II evidence
D	Directly based on Level IV evidence or extrapolated recommendations from Level I, II, or III evidenc